

CONSUMER AND PRODUCER SURPLUS
TUTORIAL SET

1. The demand curve for a product is given by $p = 50 - 2q$, where p is the price and q is the quantity demanded. If the market price is set at $p = 30$:
 - a. Find the consumer surplus.
 - b. Interpret your result.
2. The supply curve for a product is given by $p = 10 + q$, where p is the price and q is the quantity supplied. If the market price is set at $p = 40$:
 - a. Find the producer surplus.
 - b. Interpret your result.
3. The demand curve is $p = 100 - q^2$, and the supply curve is $p = 28 + q^2$. The equilibrium price and quantity are given as $q = 6$ and $p = 64$.
 - a. Find the consumer surplus.
 - b. Find the producer surplus.
4. The demand curve is $p = 150 - 3q$, and the supply curve is $p = 30 + 2q$.
 - a. Find the equilibrium price and quantity.
 - b. Calculate the consumer surplus.
 - c. Calculate the producer surplus.
 - d. Find the total surplus and explain its significance.
5. Suppose the demand for a product is given by $p = 80 - 5q$, and the supply is $p = 10 + 3q$. A government subsidy decreases the market price to $p = 20$.
 - a. Find the new equilibrium quantity after the subsidy.
 - b. Calculate the new consumer surplus.
 - c. Calculate the new producer surplus.
 - d. Compare the total surplus before and after the subsidy.